Open source Java implementation of the Beacon v2 API

D. Repchevsky [1,2], S. Capella-Gutierrez [1,2], J.L. Gelpi [1,2,3]

[1] Barcelona Supercomputing Center (BSC), Barcelona Spain.
[2] Spanish National Bioinformatics Institute - Instituto Nacional de Bioinformatica (INB-ISCIII), Barcelona Spain.
[3] Universitat de Barcelona (UB), Barcelona Spain.

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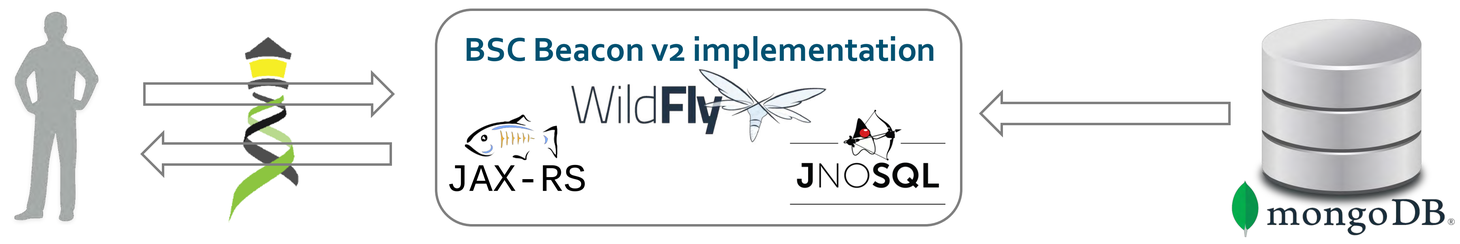
Recently approved by The Global Alliance for Genomics and Health (GA4GH) Beacon v2 introduced many changes to the protocol and the data model. The specification is split in two blocks - the Beacon v2 Framework that ensembles the protocol parts and Beacon v2 data model that provides default schemas for entities like "individual", "biosample", "sequence analysis", etc. The data model was developed in conjunction with other standards such as Phenopackets and Variation Representation Specification and is oriented to the clinical research domain although the framework is model agnostic and other models could be used. The Beacon v2 protocol underwent many enhancement with support of query filters, pagination, improved metadata, enhanced response types, etc.

BSC Beacon v2 Java implementation is based on the Jakarta EE (Enterprise Edition) and uses MongoDB database as a NoSQL backend.

Implementation:



Beacon v2 interaction

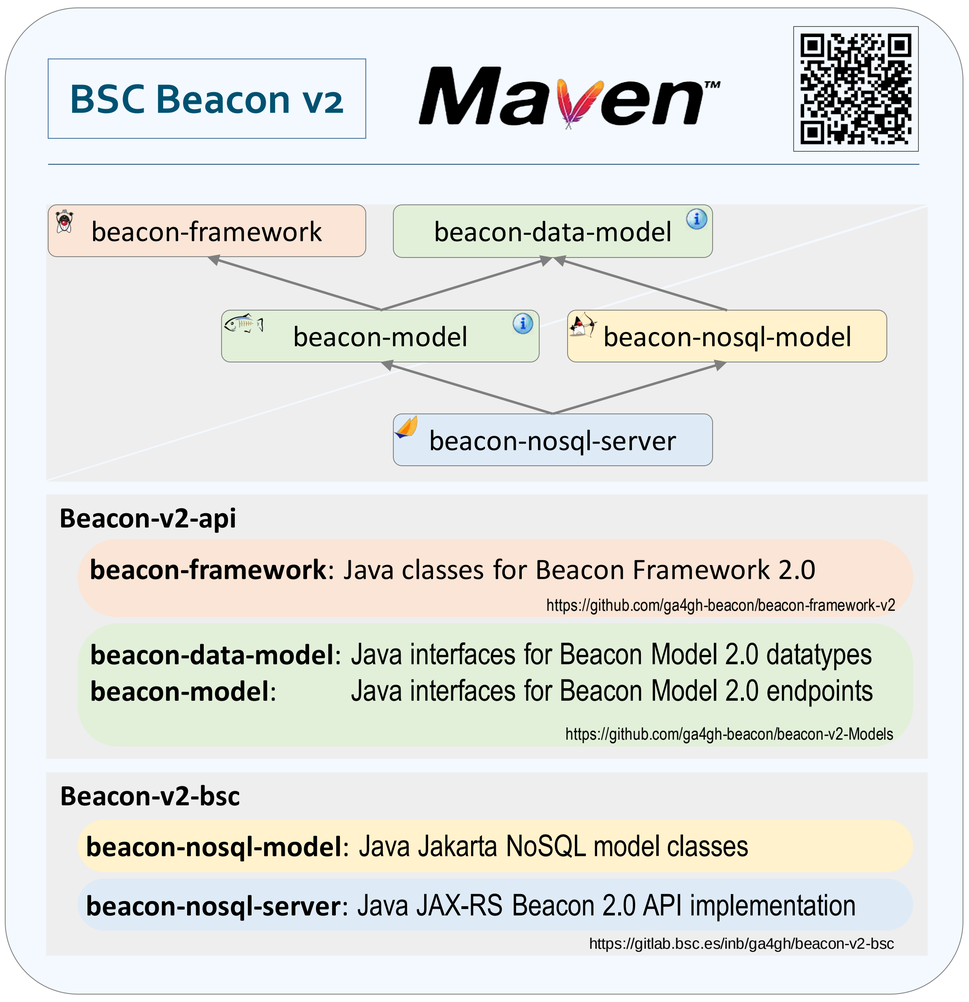
The implementation is divided in two parts:

- beacon-v2-api - Beacon v2 framework and data model Java JAX-RS / JSON-B interfaces and model classes.
- beacon-v2-bsc - JAX-RS endpoints implementation with JNoSQL data model.

The beacon-v2-api is totally abstract as it contains no implementation code and provides Java interfaces and model classes. The beacon-v2-bsc contains JNoSQL mappings for the Beacon v2 data model and an implementation of the Beacon v2 REST API interfaces provided in the beacon-v2-api.

The project uses Apache Maven build system what should simplify code usage by institutions that want to adopt it for their Beacon v2 development.

In addition to the Beacon v2 implementation, there is a simple Docker Compose script which allows prompt demo Beacon v2 deployment.



Live Demo:

https://beacons.bsc.es/beacon/v2.0.0/g_variants?geneid=NCAPH2


```
{ "response": {
    "resultSets": [ {
        "exists": true,
        "results": [ {
            "alternateBases": "G",
            "caseLevelData": [
              {"biosampleId": "NA19023"}, ...
            ],
            "identifiers": { "genomicHGVSId": "22:g.50941700C>G" },
            "molecularAttributes": {
                "geneIds": ["LMF2", "NCAPH2"],
                "molecularEffects": [ {
                  "id": "ENSGLOSSARY:0000159",
                  "label": "3_prime_UTR_variant" }
                ]
            },
            "position": {
              "assemblyId": "hs37",
              "end": [ 50941700 ],
              "refseqId": "22",
              "start": [ 50941699 ]
            },
            "referenceBases": "C",
            "variantInternalId": "chr22_50941700_C_G",
            "variantType": "SNP"
          } ...
```


The Docker Compose git repository contains synthetic cohorts datasets provided by the CINECA project. The datasets are compressed with lzma compressor. Note that compressed genomic variation data is 269 Mb big, while uncompressed it occupies almost 12 Gb. The script that initializes the database decompresses it on-the-fly, but it is still needed around 2 Gb of disk space.

Initial MongoDB Docker machine initialization may take quite a long time, but consequent runs wont reinitialize the dataset import.

The beacon-nosql-server application is downloaded from the BSC GitLab maven repository and automatically updated as part of Continuous Integration (CI) process.

The Docker Compose script is configured to bind the Beacon to the port standard http port with beacon/v2.0.0 URI.

CONTACTS:
Dmitry Repchevsky
Research Software Engineer
dmitry.repchevski@bsc.es

Spanish National Bioinformatics Institute

Barcelona Supercomputing Center
Centro Nacional de Supercomputación

@ELIXIREurope /company/elixir-europe